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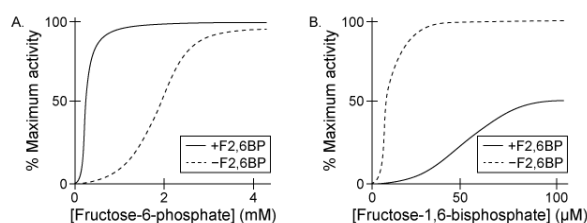


Figure 1 Activities of (A) PFK-1 and (B) F1,6BPase on their respective substrates in the presence (solid lines) and absence (dashed lines) of F2,6BP

A bifunctional enzyme that contains a phosphofructokinase-2 (PFK-2) domain and a fructose-2,6-bisphosphatase (F2,6BPase) domain controls F2,6BP levels in the liver. The PFK-2 domain converts F6P to F2,6BP, and the F2,6BPase domain converts F2,6BP back to F6P. When blood glucose levels are low, the enzyme becomes phosphorylated. This phosphorylation event simultaneously activates the F2,6BPase domain and inactivates the PFK-2 domain. Under high blood glucose conditions, the enzyme becomes dephosphorylated, activating the PFK-2 domain and inactivating the F2,6BPase domain.

All of the following statements about PFK-1 and F1,6BP are correct EXCEPT:

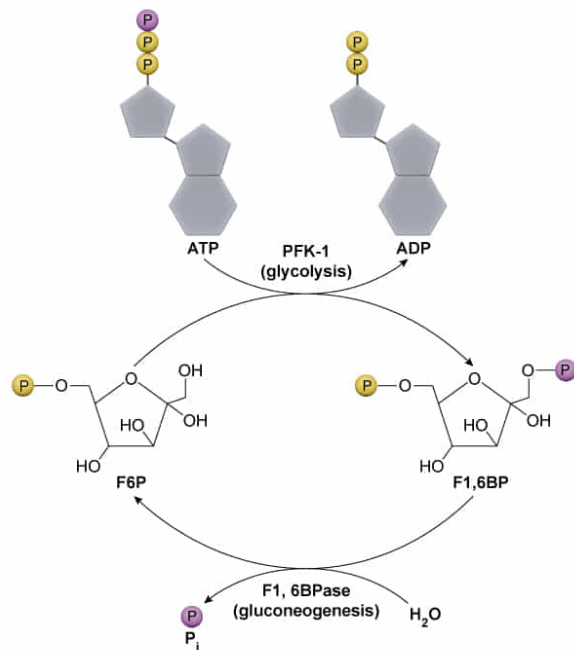
- ☐ A. PFK-1 cooperatively converts F6P to F1,6BP. [8%]
- ☐ B. F1,6BPase cooperatively converts F1,6BP to F6P. [15%]
- ☐ C. Both PFK-1 and F1,6BPase have multiple active sites for catalysis. [31%]
- ☒ D. F1,6BPase catalyzes the reverse reaction of PFK-1 to regenerate ATP. [44%]

Correct

44% answered correctly

Explanation:

PFK-1 and F1,6BPase catalyze distinct reactions



Metabolic pathways consist of both **reversible and irreversible reactions**, each of which is catalyzed by an enzyme. For a **reversible process**, the relevant enzyme catalyzes (ie, decreases the activation energy of) both the forward *and* the reverse reactions. In *irreversible* processes, the reverse reaction cannot occur, so converting the product of an irreversible reaction back to its precursor requires a separate process known as a **bypass reaction**. Bypass reactions are catalyzed by *distinct* enzymes.

Glycolysis, the conversion of glucose to pyruvate, consists of *seven* reversible reactions and *three* irreversible reactions. Gluconeogenesis converts pyruvate back to glucose, and for the reversible steps it uses the same enzymes (eg, **3-phosphoglycerate kinase**) as glycolysis to catalyze the reverse reactions.

The question asks for the incorrect statement. To **bypass the irreversible steps** of glycolysis, gluconeogenesis uses **different enzymes** than those used in glycolysis to catalyze **alternate reactions**. According to the passage, PFK-1 transfers a phosphate from ATP to F6P during glycolysis to form ADP and F1,6BP. The reverse reaction would transfer the phosphate back to ADP, regenerating ATP and F6P. However, this reaction is too thermodynamically unfavorable to occur. Instead, gluconeogenesis uses F1,6BPase to **hydrolyze** the phosphate group from F1,6BP, forming F6P and an *inorganic phosphate* (P_i), *not* ATP. Therefore, contrary to the statement in Choice D, **F1,6BPase does *not* catalyze the reverse reaction of PFK-1.**

(Choices A, B, and C) Figure 1 shows that both PFK-1 and F1,6BP have *sigmoidal* (S-shaped) *activity curves*, indicating that both enzymes are cooperative. **Cooperativity** occurs when the binding of a substrate to one active site increases the affinity of other active sites for the substrate. In the absence of cooperativity, the shapes of the curves would be **hyperbolic**. Therefore, by definition, cooperative enzymes must have multiple active sites.

Educational objective:

Metabolic pathways consist of both reversible and irreversible reactions. Opposing metabolic processes such as glycolysis and gluconeogenesis typically use the same enzymes for reversible reactions (going in opposite directions) but must use different enzymes to catalyze distinct reactions for the irreversible steps.

Time spent:0 sec**Subject:**
Biochemistry**QID:**402651**System:**
Metabolic Reactions

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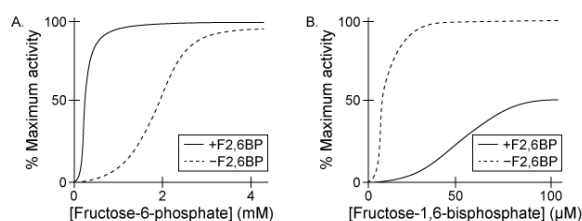


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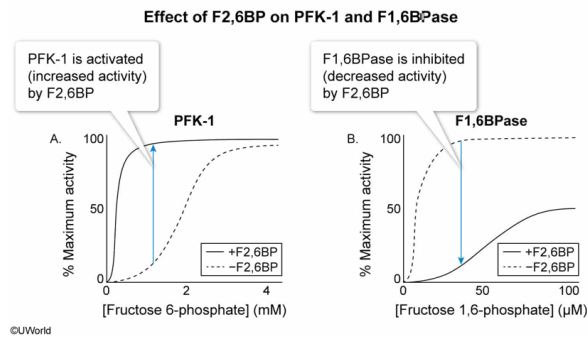
Given the information in Figure 1, are PFK-1 and F1,6BPase activated or inhibited by F2,6BP?

- ☐ A. PFK-1 is activated, and F1,6BPase is activated. [4%]
- ☒ B. PFK-1 is activated, and F1,6BPase is inhibited. [78%]
- ☐ C. PFK-1 is inhibited, and F1,6BPase is activated. [15%]
- ☐ D. PFK-1 is inhibited, and F1,6BPase is inhibited. [2%]

Correct

77% answered correctly

Explanation:



Allosteric effectors are small molecules that **bind to enzymes** at sites other than the active site. Upon binding, allosteric effectors induce conformational changes in enzymes that **alter enzyme activity**. **Allosteric activators increase** enzymatic activity, whereas **allosteric inhibitors decrease** enzymatic activity. Many **metabolic pathways** are regulated by allosteric effectors, and an effector that **activates one metabolic pathway** often **inhibits the opposing pathway** (eg, F2,6BP in glycolysis and gluconeogenesis).

Figure 1 shows the rates of the reactions catalyzed by PFK-1 and F1,6BPase, respectively, with the substrate concentrations for each shown on the corresponding x-axis. Part A shows that in the presence of F2,6BP, the glycolysis enzyme PFK-1 converts its substrate, F6P, into product at a *higher* rate than in the absence of F2,6BP. Conversely, the gluconeogenesis enzyme F1,6BPase (shown in part B) converts its substrate, F1,6BP, into product at a *lower* rate in the presence of F2,6BP than in its absence. Therefore, **PFK-1 is activated** by F2,6BP and **F1,6BPase is inhibited**.

Educational objective:

Metabolic pathways are commonly regulated by allosteric effectors. Allosteric effectors bind enzymes at sites other than the active site and induce conformational changes that alter enzymatic activity. A molecule that allosterically activates one pathway often allosterically inhibits the opposing pathway.

Time spent: 0 sec

Subject:
Biochemistry

QID: 402652

System:
Metabolic Reactions

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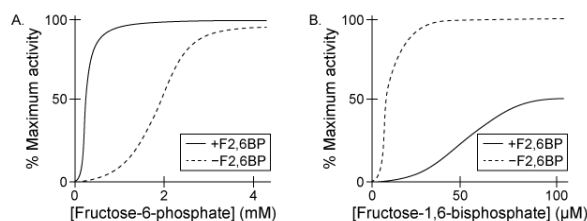


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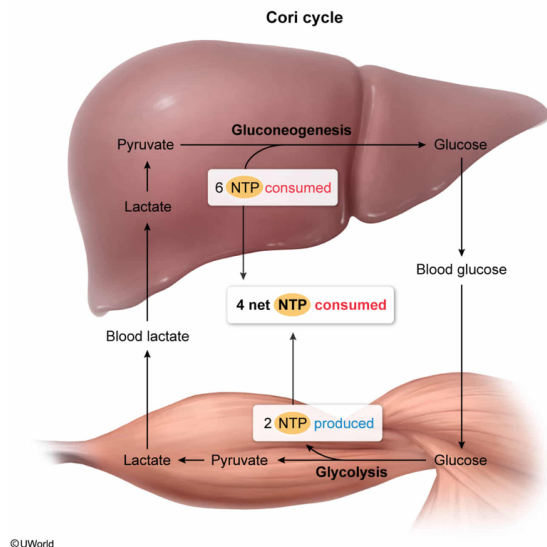
If one glucose molecule is converted to pyruvate in the muscle and then the resulting pyruvate molecules are converted back to glucose in the liver, what is the net number of NTPs consumed in this process?

- ☐ A. 2 [13%]
- ☒ B. 4 [38%]
- ☐ C. 6 [28%]
- ☐ D. 8 [19%]

Correct

39% answered correctly

Explanation:



Metabolism consists of both **catabolic and anabolic** pathways. **Catabolic pathways** degrade complex molecules into simpler molecules to **produce high-energy nucleotide triphosphates** (ie, NTPs such as ATP or GTP). NTPs provide energy for many biological processes, including **muscle contraction**. **Anabolic pathways** use simple molecules (eg, amino acids, sugars) as precursors to **synthesize more complex molecules** (eg, proteins, polysaccharides). **Anabolism requires energy input** from NTPs. Most catabolic processes are paired with an **opposing anabolic process**.

Net NTP production or consumption refers to the **difference** between the numbers of NTPs **produced** and **consumed** in a process. For example, **glycolysis** (a **catabolic process**) consumes 2 ATPs and produces 4 ATPs per glucose molecule; therefore, glycolysis has a *net* production of 2 ATPs ($4 - 2 = 2$) per glucose. In contrast, the passage states that **gluconeogenesis** (an **anabolic process**) *consumes* a net total of 6 NTPs (4 ATPs and 2 GTPs) to produce one glucose molecule.

Glycolysis in the muscles is connected to gluconeogenesis in the liver by the **Cori cycle**. In this scenario, the 2 net NTPs that are produced when glucose becomes pyruvate (glycolysis) partially offset the 6 net NTPs that are later consumed when pyruvate is converted back to glucose (gluconeogenesis). Therefore, the *net* number of NTPs consumed by the process is $6 - 2 = 4$ **net NTPs consumed** in one round of the Cori cycle. Alternatively, this may be described as 8 total NTPs consumed (2 in glycolysis and 6 in gluconeogenesis) and 4 total NTPs produced (in glycolysis only). $8 - 4 = 4$ **net NTPs consumed**.

(Choice A) Glycolysis *produces* 2 NTPs per glucose molecule, but these molecules, as well as 4 other NTPs, are effectively *consumed* by gluconeogenesis.

(Choice C) Gluconeogenesis consumes a *total* of 6 NTPs, but in the given scenario glycolysis also *produces* 2 NTPs, so the *net* result is the consumption of 4 NTPs.

(Choice D) The *total* number of NTPs consumed by one round of the Cori cycle is 8 (2 from glycolysis and 6 from gluconeogenesis), but the question asks for the *net* number (ie, total consumed – total produced).

Educational objective:

Catabolic processes lead to energy production, typically in the form of ATP equivalents. In contrast, anabolic processes consume energy. Glycolysis (a catabolic process) and gluconeogenesis (an anabolic process) are connected by the Cori cycle.

Time spent:0 sec

Subject:
Biochemistry

QID:402653

System:
Metabolic Reactions

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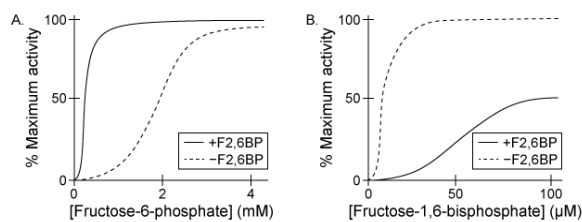


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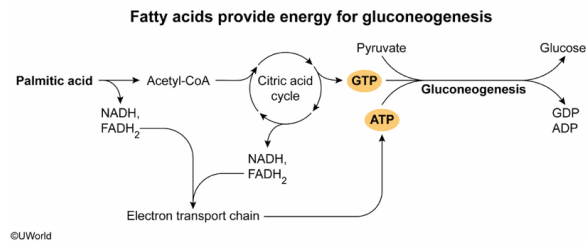
Which metabolic process most likely provides the energy necessary for sustained gluconeogenesis?

- ☒ A. Fatty acid oxidation [54%]
- ☐ B. Glycogenolysis [20%]
- ☐ C. Fermentation [8%]
- ☐ D. Pentose phosphate pathway [16%]

Correct

54% answered correctly

Explanation:



Gluconeogenesis is an **anabolic process** that synthesizes glucose from smaller precursors such as pyruvate. This pathway is used to increase blood glucose levels during times of fasting. According to the passage, gluconeogenesis requires **energy input** in the form of **6 ATP equivalents** (ATP or GTP) for each glucose molecule synthesized. The energy required to produce these ATP equivalents must be **provided by a catabolic process**. Because the result of gluconeogenesis is the *synthesis* of carbohydrates, it is unlikely that the necessary energy is provided by carbohydrate *degradation*. Instead, the needed ATP equivalents are typically produced by **fatty acid oxidation**.

Fatty acid degradation to acetyl-CoA yields NADH and FADH₂, both of which can enter the **electron transport chain** to produce ATP. In addition, the resulting acetyl-CoA can enter the **citric acid cycle** to produce more NADH and FADH₂, as well as GTP (an ATP equivalent). Therefore, of the choices given, **fatty acid oxidation** most likely **provides the energy needed** for sustained gluconeogenesis.

(Choice B) **Glycogenolysis** produces glucose, so when glycogen is present, gluconeogenesis is *not* required. Accordingly, gluconeogenesis generally occurs *after* glycogen stores have been depleted, so glycogenolysis would not be able to provide the needed energy.

(Choice C) In most eukaryotes, fermentation is the conversion of pyruvate to lactate. The resulting lactate can enter gluconeogenesis after being converted back to pyruvate but does not provide the needed energy.

(Choice D) The **pentose phosphate pathway** converts glucose and glycolysis intermediates to ribose-5-phosphate. This process neither consumes nor produces any ATP, so it does not provide the energy for gluconeogenesis.

Educational objective:

Gluconeogenesis is an anabolic process that requires energy input from ATP equivalents. The necessary energy is provided by catabolic processes, most commonly by fatty acid oxidation.

Time spent: 0 sec
Subject:
Biochemistry

QID: 402654
System:
Metabolic Reactions

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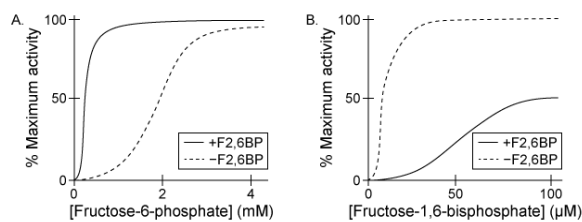


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The information in the passage and the data in Figure 1 support the conclusion that when glycolysis is active, the bifunctional PFK-2/F2,6BPase enzyme is:

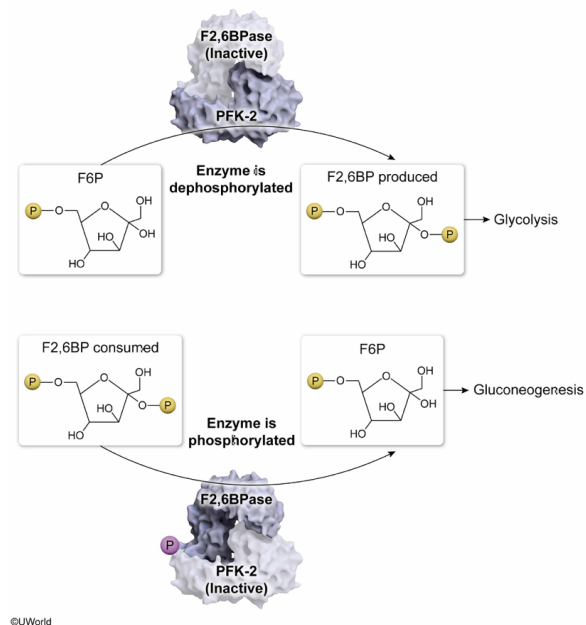
- ☐ A. phosphorylated and consumes F2,6BP. [13%]
- ☐ B. phosphorylated and produces F2,6BP. [16%]
- ☐ C. dephosphorylated and consumes F2,6BP. [21%]
- ☒ D. dephosphorylated and produces F2,6BP. [48%]

Correct

47% answered correctly

Explanation:

Enzyme phosphorylation activates F2,6BPase and inactivates PFK-2



The **rate of glycolysis** is largely **controlled by PFK-1** activity. When PFK-1 is active, glycolysis proceeds quickly, whereas when PFK-1 activity is downregulated, glycolysis is slow. **PFK-1 is tightly regulated** by several allosteric effectors, including the **inhibitors ATP and citrate** and the **activators ADP, AMP, and F2,6BP**.

The data in Figure 1 confirm that PFK-1 is *most* active when F2,6BP is present, indicating that glycolysis is most active when F2,6BP is abundant. Therefore, **conditions that promote glycolysis** most likely also **promote F2,6BP synthesis**. Because the **PFK-2** domain *produces* F2,6BP, it must be *active* during glycolysis. The passage states that the PFK-2 domain of the bifunctional enzyme is active (and the F2,6BPase domain is inactive) when it is dephosphorylated. Therefore, when glycolysis is active, the PFK-2/F2,6BPase enzyme is most likely **dephosphorylated** and **produces F2,6BP**.

(Choice A) When the PFK-2/F2,6BPase enzyme is phosphorylated, the F2,6BPase domain is active and consumes F2,6BP. However, this occurs when *gluconeogenesis*, not glycolysis, is active.

(Choice B) If PFK-2/F2,6BPase were phosphorylated, it would *consume* F2,6BP, converting it to F6P, and stimulate *gluconeogenesis* rather than glycolysis.

(Choice C) When PFK-2/F2,6BPase is dephosphorylated, it *produces* F2,6BP to stimulate glycolysis, rather than consumes F2,6BP.

Educational objective:

The rate of glycolysis is controlled by PFK-1, which is stimulated by the allosteric effector F2,6BP. When glycolysis is active, F2,6BP synthesis is also most likely active.

Time spent: 0 sec
Subject:
Biochemistry

QID: 402655
System:
Metabolic Reactions

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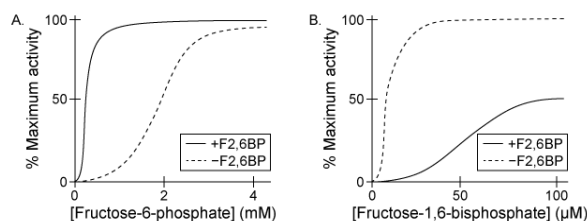


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What effect does glucagon most likely have on F2,6BP synthesis in the liver?

Glucagon:

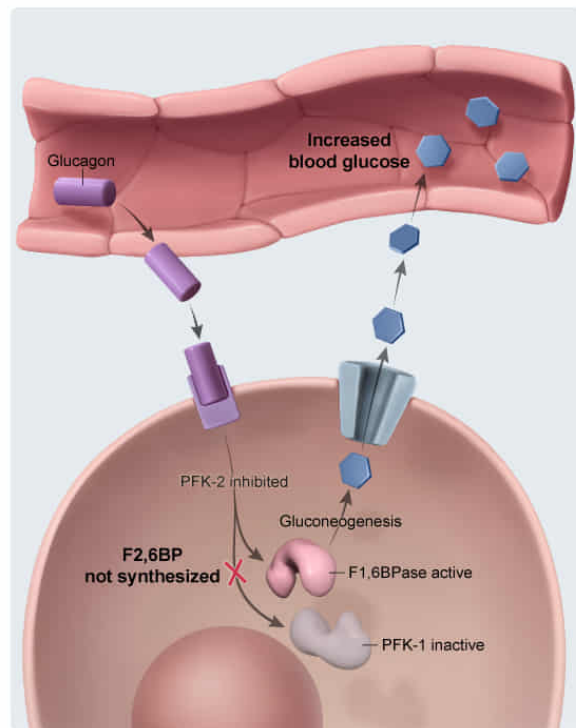
- ☐ A. does not alter F2,6BP synthesis. [4%]
- ☐ B. causes an increase in F2,6BP synthesis. [30%]
- ☒ C. causes a decrease in F2,6BP synthesis. [61%]
- ☐ D. causes F2,6BP to be synthesized and degraded in a cycle. [3%]

Correct

61% answered correctly

Explanation:

Effect of glucagon on F2,6BP synthesis



Blood glucose levels are controlled primarily by the peptide hormones **insulin** and **glucagon**. When **blood glucose** levels are **high**, the **pancreas releases insulin**, which induces glucose uptake into cells. Insulin also induces **glycolysis** and **glycogen synthesis**, helping remove glucose from the bloodstream. Conversely, when **blood glucose** levels are **low**, the **pancreas releases glucagon**. When glucagon binds to receptors on liver cells, it induces an **intracellular response** that ultimately upregulates **glycogenolysis** and **gluconeogenesis**, leading to the production of glucose that can be released into the blood.

According to Figure 1, F2,6BP stimulates PFK-1 activity to increase glycolysis and inhibits F1,6BPase activity to decrease gluconeogenesis. Glucagon has the opposite effect, *decreasing* the rate of glycolysis and *increasing* the rate of gluconeogenesis. Therefore, glucagon most likely causes decreased PFK-2 activity, resulting in **decreased F2,6BP synthesis**. This downregulates glycolysis and upregulates gluconeogenesis. These effects ultimately help to increase blood glucose levels.

(Choice A) If glucagon had no effect on F2,6BP synthesis, it would be *ineffective* in regulating glycolysis and gluconeogenesis.

(Choice B) Increased F2,6BP synthesis would lead to *increased* glycolysis and *decreased* gluconeogenesis. Glucagon has the *opposite* effect.

(Choice D) Glucagon causes an intracellular response that *increases* F2,6BP degradation and *decreases* synthesis. It does *not* promote a cycle of synthesis and degradation.

Educational objective:

The hormones insulin and glucagon regulate blood glucose levels. When blood glucose is high, the pancreas releases insulin, which upregulates glucose uptake, glycolysis, and glycogen synthesis. When blood glucose is low, the pancreas releases glucagon, which stimulates gluconeogenesis and glycogenolysis to release glucose from the liver into the bloodstream.

Time spent:0 sec

Subject:
Biochemistry

QID:402656

System:
Metabolic Reactions